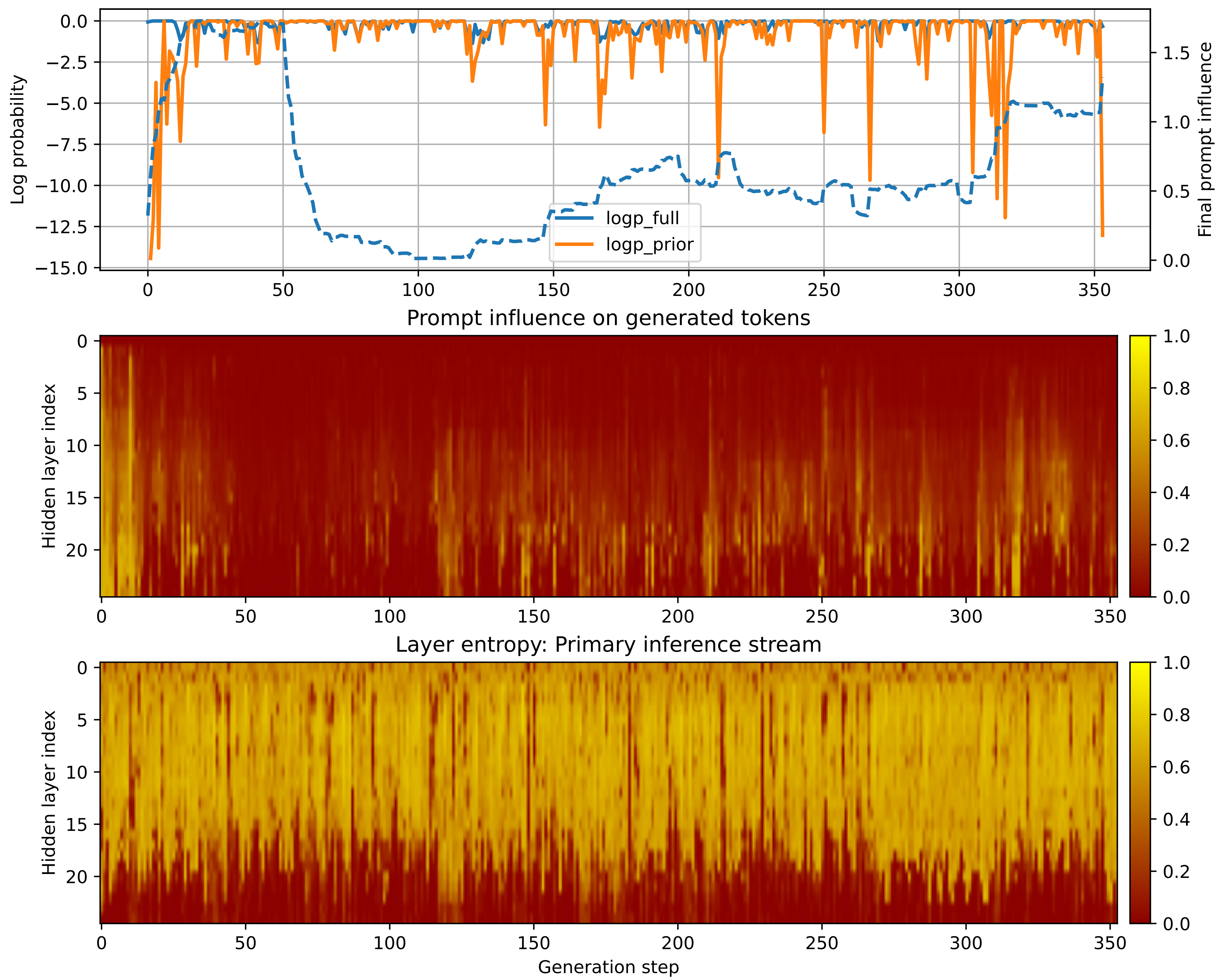
Prompt 1

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PROMPT:
Task ID: PYC069  Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests.  Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC069  Your code must contain this required symbol:
def knapsack_01  Implement this task: define
function knapsack_01(weights, values, capacity)..  Start your
answer with the check-token line, then provide
the Python implementation.

MODEL RESPONSE:
# CHECK_TOKEN: knapsack_01  # Function definition for knapsack_01
def knapsack_01(weights, values, capacity):
# Check if the input parameters are valid     if not
isinstance(weights, list) or not isinstance(values, list)
or not isinstance(capacity, int):         raise ValueError("All
input parameters must be lists.")          #
Check if the capacity is less than or equal to 0     if capacity
< 0:         raise ValueError("The capacity
must be a non-negative integer.")          # Initialize the
maximum value and the current value     max_value
= float('-inf')     current_value = 0         # Iterate through
the weights and values     for weight in
weights:         for value in values:            # Calculate the
subproblem: find the maximum value that can
be obtained by selecting the current weight and value
sub_value = weight * value + current_value
# Check if the subproblem is feasible (i.e., the capacity is not
exceeded)             if sub_value <=
capacity:                # Update the maximum value and current
value                  max_value =
max(max_value, sub_value)                current_value =
sub_value          # Return the maximum value
return max_value  # Example usage weights = [10, 20, 30, 40]
values = [60, 80, 100, 120] capacity = 50  # Call
the knapsack_01 function max_value = knapsack_01(weights, values,
capacity) print("Maximum value:", max_value)
# Output: 150
```